Brewing Insights: Cleaning, Exploring, and Analyzing 10,000 Messy Cafe Sales Records to Drive Business Strategy

Moshood A. Garuba

2025-05-03

Project Goal:

To clean and analyze messy transactional sales data from The Dirty Cafe to uncover customer behavior, top-selling items, revenue trends, and operational inefficiencies. Ultimately, the analysis will inform data-driven decisions to increase revenue, optimize inventory, and enhance customer experience.

Narrative & Structure:

1. Introduction / Business Background:

The Dirty Cafe is a fast-growing urban café chain with multiple outlets. Recently, the café started recording digital sales, but data entry issues and inconsistent systems have introduced messiness into the dataset. As a data analyst, you're tasked with transforming the raw data into actionable business insights to support growth.

```r
# Load required libraries
library(tidyverse)
library(lubridate)
library(readr)
library(janitor)
library(dplyr)
library(kableExtra)
library(naniar)
library(lubridate)
library(dlookr)
library(patchwork)
library(gt)
library(gtsummary)
library(report)

# Read the CSV
cafe_data <- read_csv("dirty_cafe_sales.csv")
```

2. Data Cleaning:

```r
#Checking the names of the variables
names(cafe_data)
```

```
## [1] "Transaction ID"   "Item"             "Quantity"         "Price Per Unit"
## [5] "Total Spent"      "Payment Method"   "Location"         "Transaction Date"
```

```r
# Checking the structure of the variables
glimpse(cafe_data)
```

```
## Rows: 10,000
## Columns: 8
## $ `Transaction ID`   <chr> "TXN_1961373", "TXN_4977031", "TXN_4271903", "TXN_7~
## $ Item               <chr> "Coffee", "Cake", "Cookie", "Salad", "Coffee", "Smo~
## $ Quantity           <chr> "2", "4", "4", "2", "2", "5", "3", "4", "5", "5", "~
## $ `Price Per Unit`   <chr> "2.0", "3.0", "1.0", "5.0", "2.0", "4.0", "3.0", "4~
## $ `Total Spent`      <chr> "4.0", "12.0", "ERROR", "10.0", "4.0", "20.0", "9.0~
## $ `Payment Method`   <chr> "Credit Card", "Cash", "Credit Card", "UNKNOWN", "D~
## $ Location           <chr> "Takeaway", "In-store", "In-store", "UNKNOWN", "In-~
## $ `Transaction Date` <chr> "2023-09-08", "2023-05-16", "2023-07-19", "2023-04-~
```

```r
# Checking the number of rows and columns
dim(cafe_data)
```

```
## [1] 10000     8
```

```r
# Getting the head of the data
head(cafe_data) %>% kable()
```

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card	NA	2023-03-31

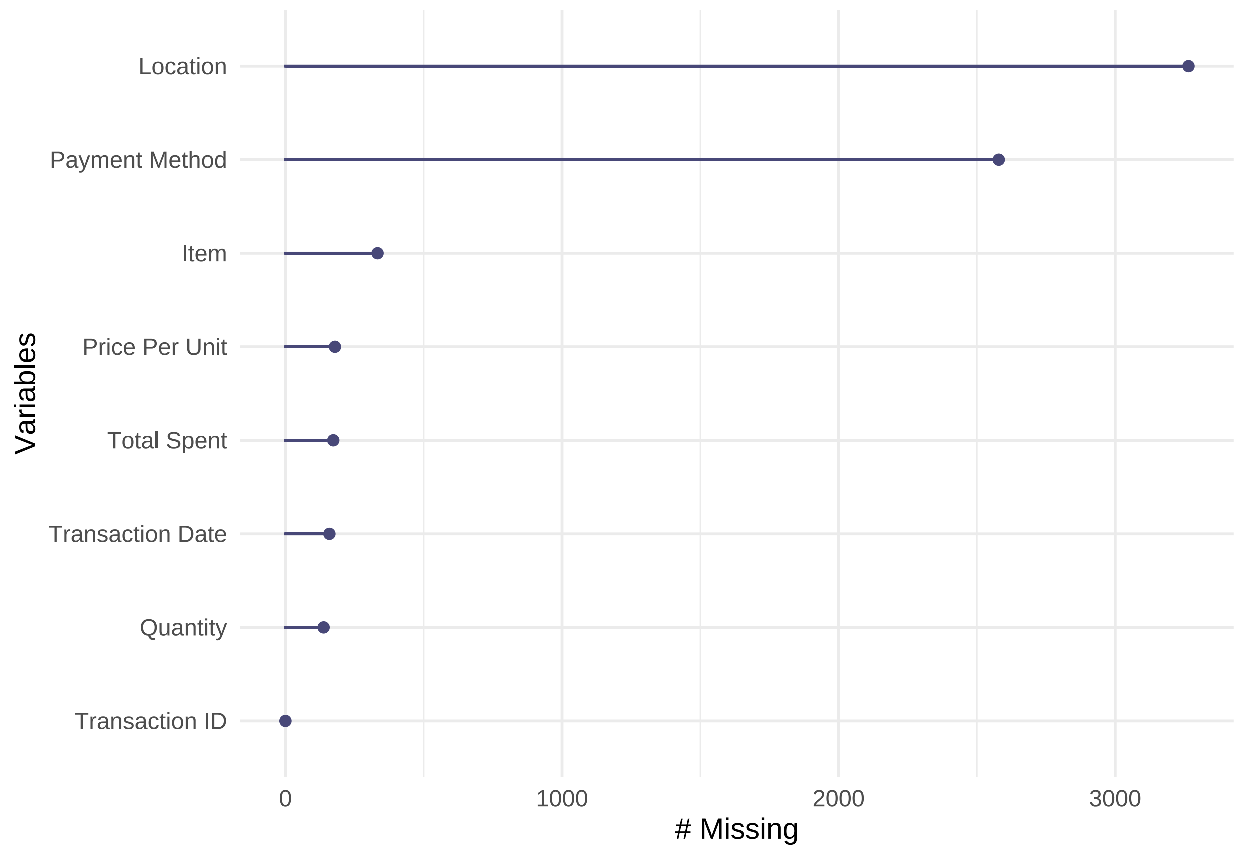
```r
# Getting the tail of the data

tail(cafe_data) %>% kable()
```

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_7851634	UNKNOWN	4	4.0	16.0	NA	NA	2023-01-08
TXN_7672686	Coffee	2	2.0	4.0	NA	UNKNOWN	2023-08-30

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_9659401	NA	3	NA	3.0	Digital Wallet	NA	2023-06-02
TXN_5255387	Coffee	4	2.0	8.0	Digital Wallet	NA	2023-03-02
TXN_7695629	Cookie	3	NA	3.0	Digital Wallet	NA	2023-12-02
TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07

```
#Visualizing missing data
gg_miss_var(cafe_data)
```




```
#Calculating the total missing data

sum(!complete.cases(cafe_data))
```

```
## [1] 5450
```

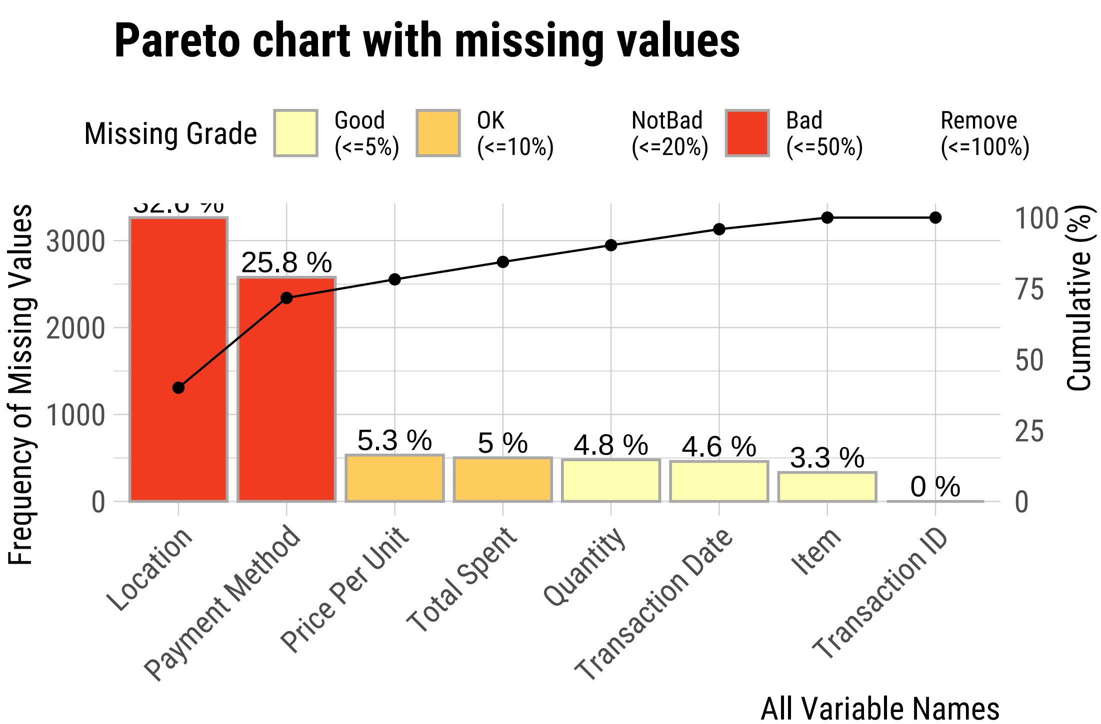
```
#Fixing data problem

cafe_data_cleaned <- cafe_data %>%
  mutate(
```

```r
    Quantity = as.numeric(Quantity),
    `Price Per Unit` = as.numeric(`Price Per Unit`),
    `Total Spent` = as.numeric(`Total Spent`),
    `Transaction Date` = as.Date(`Transaction Date`)
  ) %>%
  mutate(across(where(is.character), str_to_sentence))
```

```
## Warning: There were 3 warnings in `mutate()`.
## The first warning was:
## i In argument: `Quantity = as.numeric(Quantity)`.
## Caused by warning:
## ! NAs introduced by coercion
## i Run `dplyr::last_dplyr_warnings()` to see the 2 remaining warnings.
```

```r
plot_na_pareto(cafe_data_cleaned)
```



We have a large number of Na which is very bad strategy for us to remove them, we need to fill the NA with tag call missing so that we can make account for them

- For **Character** type of data we use missing.
- For **Numeric** type of data we will use machine learning alogrithms to fix the **NA** problem.

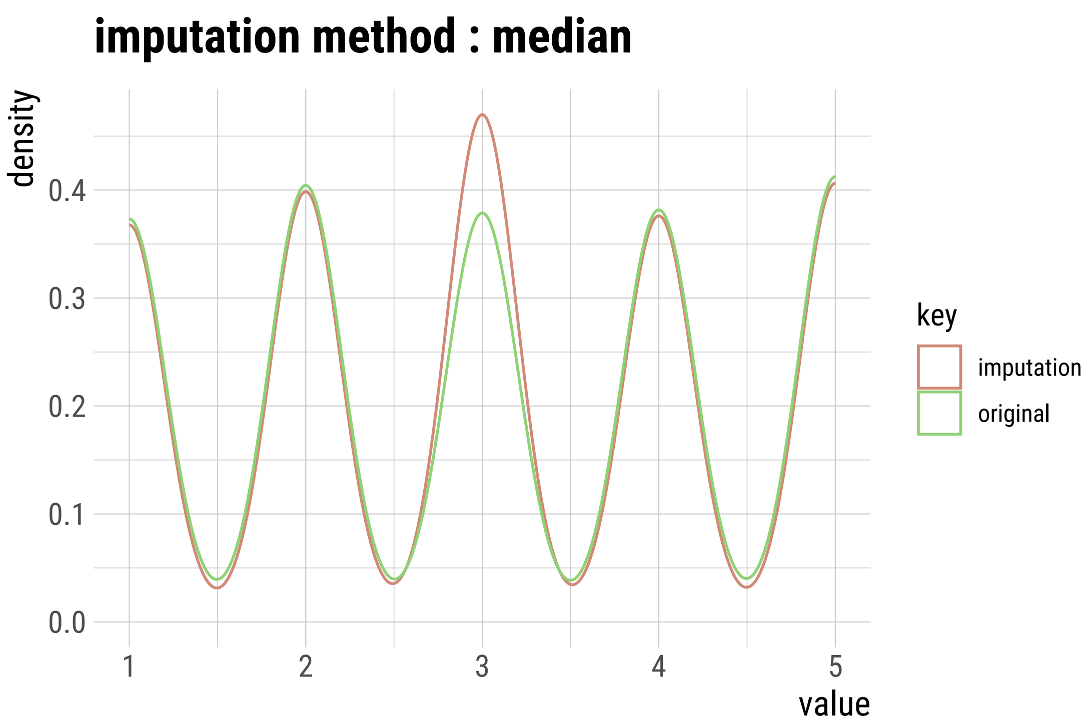
```r
#replace "Na" with the tag "missing"
cafe_data_cleaned<- cafe_data_cleaned %>%
```

```r
  mutate(across(where(is.character), ~ replace_na(., "Missing")))

str(cafe_data_cleaned)
```

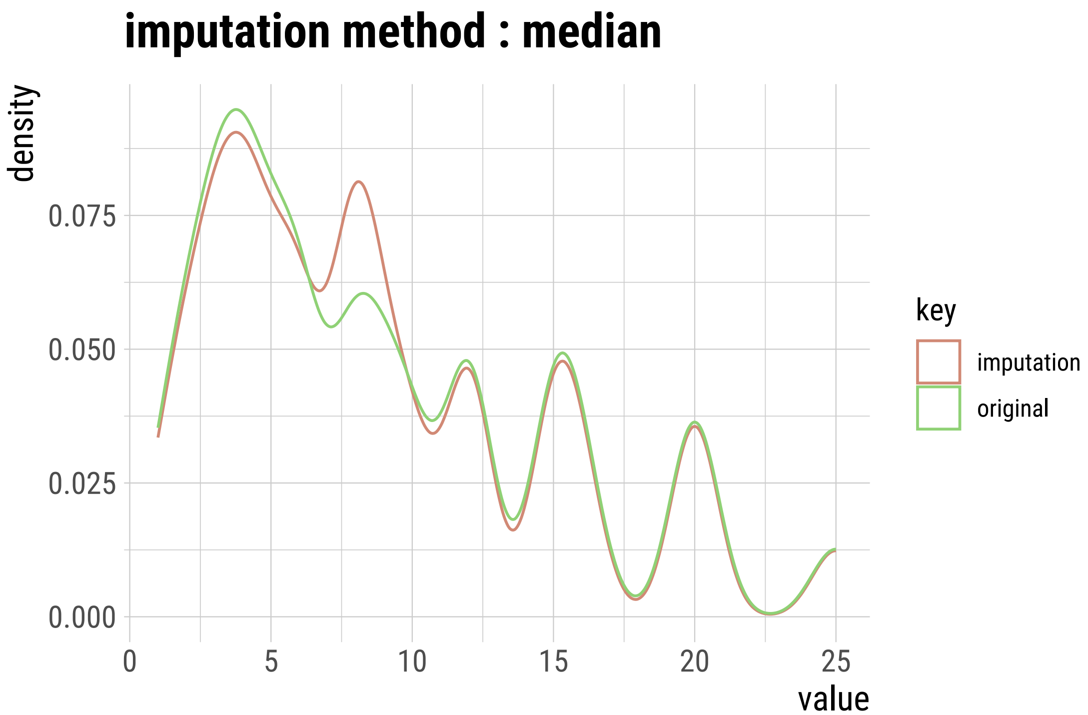
```
## tibble [10,000 x 8] (S3: tbl_df/tbl/data.frame)
##  $ Transaction ID  : chr [1:10000] "Txn_1961373" "Txn_4977031" "Txn_4271903" "Txn_7034554" ...
##  $ Item            : chr [1:10000] "Coffee" "Cake" "Cookie" "Salad" ...
##  $ Quantity        : num [1:10000] 2 4 4 2 2 5 3 4 5 5 ...
##  $ Price Per Unit  : num [1:10000] 2 3 1 5 2 4 3 4 3 4 ...
##  $ Total Spent     : num [1:10000] 4 12 NA 10 4 20 9 16 15 20 ...
##  $ Payment Method  : chr [1:10000] "Credit card" "Cash" "Credit card" "Unknown" ...
##  $ Location        : chr [1:10000] "Takeaway" "In-store" "In-store" "Unknown" ...
##  $ Transaction Date: Date[1:10000], format: "2023-09-08" "2023-05-16" ...
```

```r
# We are trying to iterate a better method for imputing the missing values for Quantity
imputate_na(cafe_data_cleaned,Quantity,method = "median") %>% plot()
```



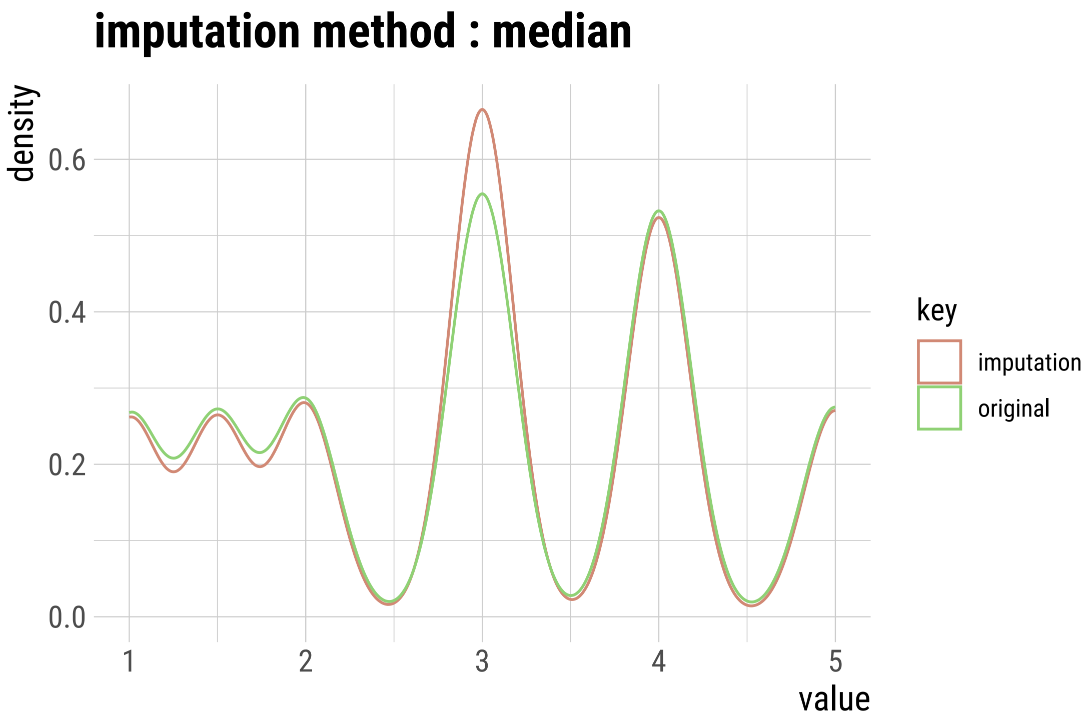

```r
#imputate_na(cafe_data_cleaned,Quantity,method = "mean") %>% plot()
#imputate_na(cafe_data_cleaned,Quantity,method = "mode") %>% plot()

#For Total spent
imputate_na(cafe_data_cleaned,`Total Spent`,method = "median") %>% plot()
```




```r
#imputate_na(cafe_data_cleaned,`Total Spent`,method = "mean") %>% plot()
#imputate_na(cafe_data_cleaned,`Total Spent`,method = "mode") %>% plot()

#For Price Per Unit
imputate_na(cafe_data_cleaned,`Price Per Unit`,method = "median") %>% plot()
```




```
#imputate_na(cafe_data_cleaned,`Price Per Unit`,method = "mean") %>% plot()
#imputate_na(cafe_data_cleaned,`Price Per Unit`,method = "mode") %>% plot()

cafe_data_cleaned<- cafe_data_cleaned %>%
  mutate(Quantity = imputate_na(cafe_data_cleaned,Quantity,method = "median"),
         `Total Spent` = imputate_na(cafe_data_cleaned,`Total Spent`,method = "median"),
         `Price Per Unit` = imputate_na(cafe_data_cleaned,`Price Per Unit`,method = "median"))
```

3. Exploratory Data Analysis (EDA)

Descriptive Analysis of Quantity, Price Per Unit and Total Spent.

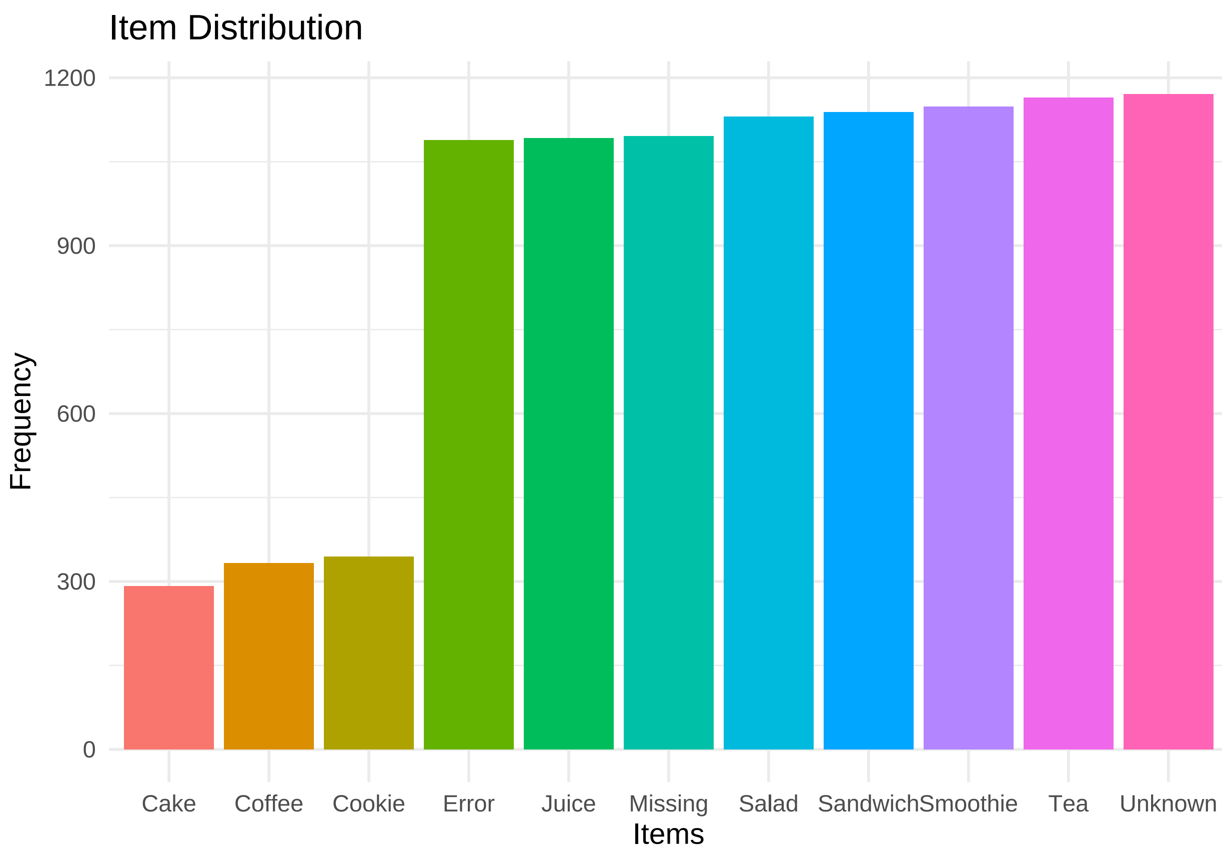
```
cafe_data_cleaned %>%
  select(Quantity, `Price Per Unit`, `Total Spent`) %>%
  summarise(across(everything(),
                   .fns = list(
                     min = ~min(.x, na.rm = TRUE),
                     median = ~median(.x, na.rm = TRUE),
                     average = ~mean(.x, na.rm = TRUE)
                   ))) %>%
  pivot_longer(cols = everything(),
               names_to = c("Variable", "Statistic"),
```

```
               names_sep = "_",
               values_to = "Value") %>%  kable()
```

Variable	Statistic	Value
Quantity	min	1.00000
Quantity	median	3.00000
Quantity	average	3.02710
Price Per Unit	min	1.00000
Price Per Unit	median	3.00000
Price Per Unit	average	2.95265
Total Spent	min	1.00000
Total Spent	median	8.00000
Total Spent	average	8.87795

What are the Item distribution?

```
cafe_data_cleaned %>%
  group_by(Item) %>%
  summarise(n = n()) %>%
  ggplot(aes(x = Item, y = sort(n)))+
  geom_bar(stat = "Identity", aes(fill = Item),
           show.legend = F)+
  labs(title = "Item Distribution", x = "Items", y = "Frequency")+
  theme_minimal()
```



What are the items sold by quantity and revenue?

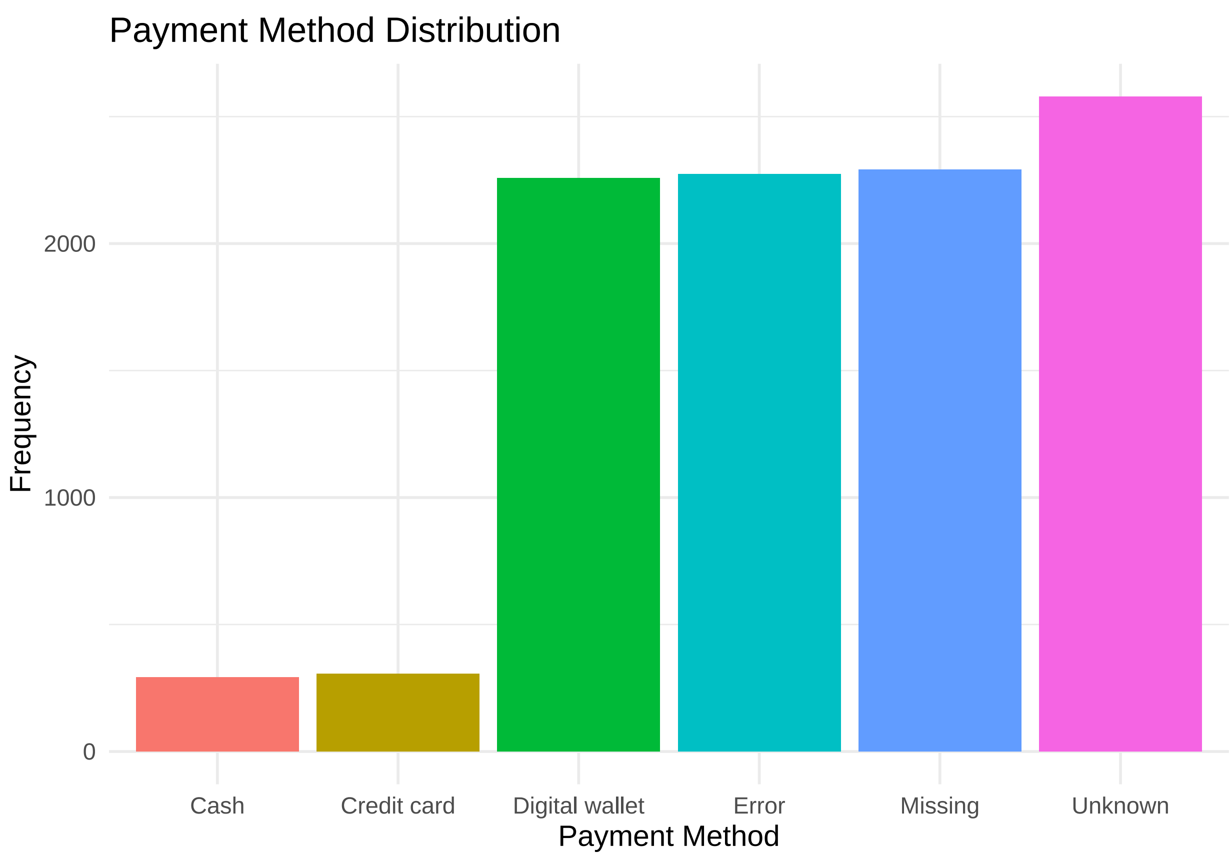
```r
cafe_data_cleaned %>%
  group_by(Item) %>%
  summarise(
    `Total Quantity` = sum(Quantity, na.rm = TRUE),
    `Total Revenue` = sum(`Total Spent`, na.rm = TRUE)
  ) %>%
  top_n(n = 10, wt = `Total Revenue`) %>% kable()
```

Item	Total Quantity	Total Revenue
Cake	3467	10341.0
Coffee	3551	7184.0
Cookie	3249	3526.0
Juice	3514	10432.0
Missing	1016	3017.0
Salad	3469	17021.0
Sandwich	3428	13484.0
Smoothie	3353	13132.0
Tea	3319	5119.5
Unknown	1006	2933.5

```r
cafe_data_cleaned %>%
  group_by(`Payment Method`) %>%
  summarise(n = n()) %>%
  arrange(n) %>% kable()
```

Payment Method	n
Unknown	293
Error	306
Cash	2258
Credit card	2273
Digital wallet	2291
Missing	2579

```r
cafe_data_cleaned %>%
  group_by(`Payment Method`) %>%
  summarise(n = n()) %>%
  ggplot(aes( x = `Payment Method`, y = sort(n)))+
  geom_bar(aes(fill = `Payment Method`), stat = "Identity", show.legend = F)+
  labs(title = "Payment Method Distribution", y = "Frequency")+
  theme_minimal()
```



Total Amount Spent

```
cafe_data_cleaned %>%
  group_by(Item) %>%
  summarise(n = n(),
            Total = sum(`Total Spent`),
            `Average Amount Spent` = round(mean(`Total Spent`),2)) %>%
  arrange(n) %>% kable()
```

Item	n	Total	Average Amount Spent
Error	292	2589.5	8.87
Missing	333	3017.0	9.06
Unknown	344	2933.5	8.53
Tea	1089	5119.5	4.70
Cookie	1092	3526.0	3.23
Smoothie	1096	13132.0	11.98
Sandwich	1131	13484.0	11.92
Cake	1139	10341.0	9.08
Salad	1148	17021.0	14.83
Coffee	1165	7184.0	6.17
Juice	1171	10432.0	8.91

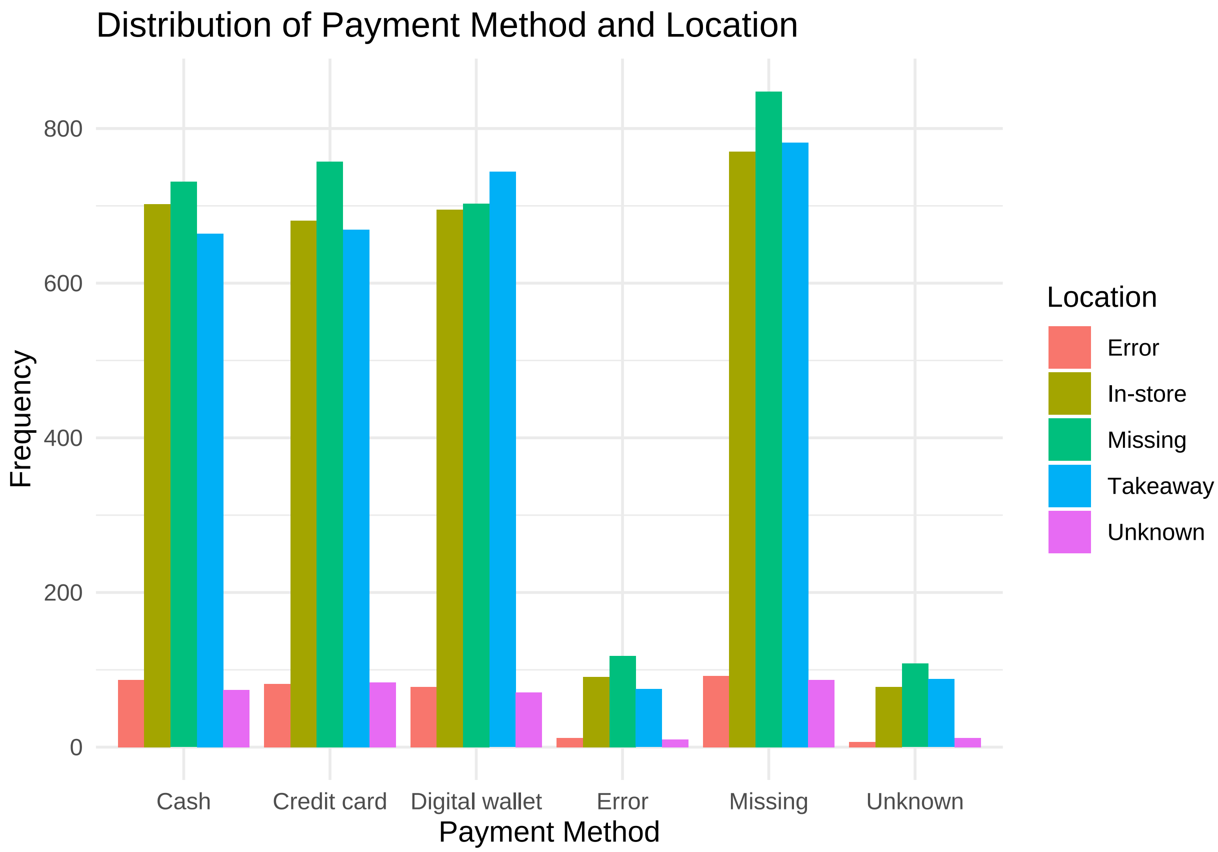
```
cafe_data_cleaned %>%
  group_by(Location) %>%
  summarise(n = n(),
            `Total Quantity` = sum(Quantity),
            `Average Quantity` = round(mean(Quantity),2)) %>% kable()
```

Location	n	Total Quantity	Average Quantity
Error	358	1100	3.07
In-store	3017	9139	3.03
Missing	3265	9893	3.03
Takeaway	3022	9176	3.04
Unknown	338	963	2.85

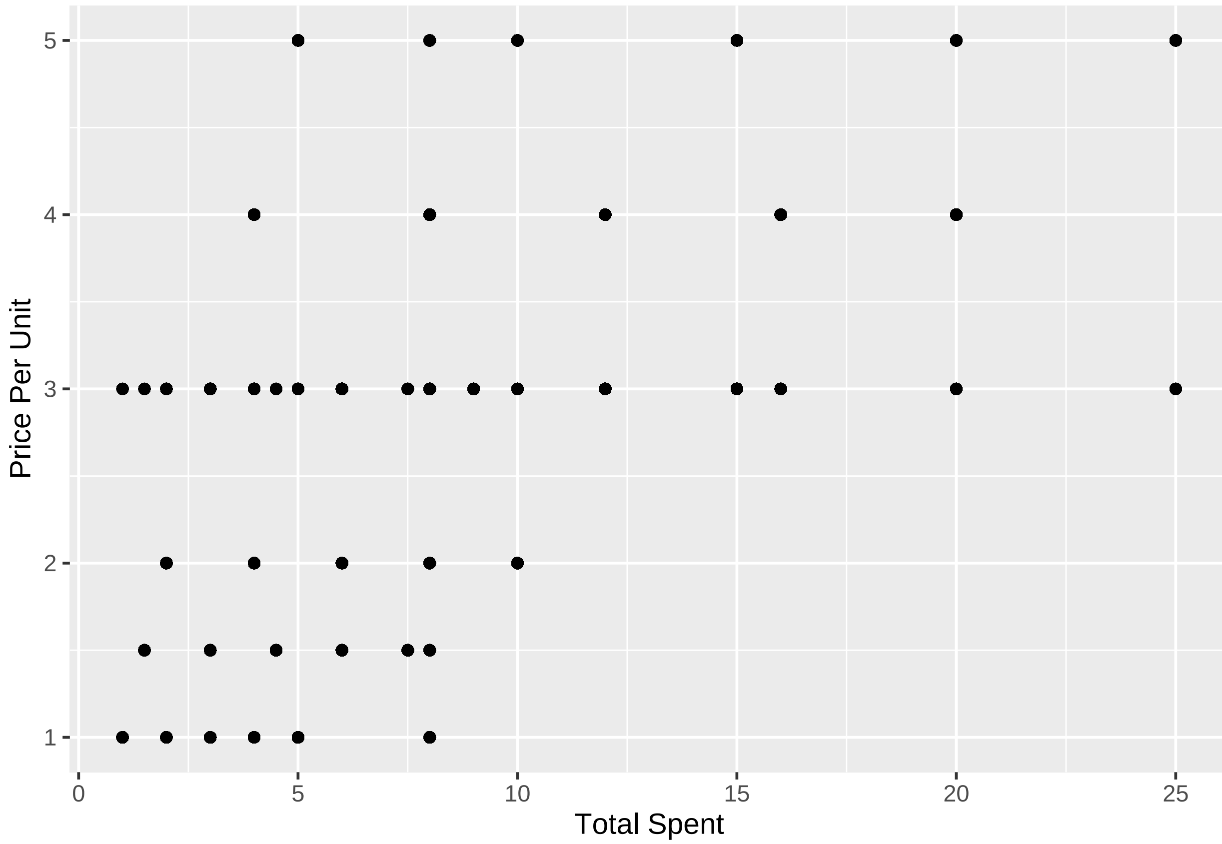
```
cafe_data_cleaned %>%
  group_by(`Payment Method`,Location) %>%
  summarise(n = n()) %>%
  ggplot(aes( x= `Payment Method`, y = n))+
  geom_bar(aes(fill = Location), stat = "identity",
          position = "dodge")+
```

```
  labs(title = "Distribution of Payment Method and Location",
       y = "Frequency")+
  theme_minimal()
```

```
## `summarise()` has grouped output by 'Payment Method'. You can override using
## the `.groups` argument.
```



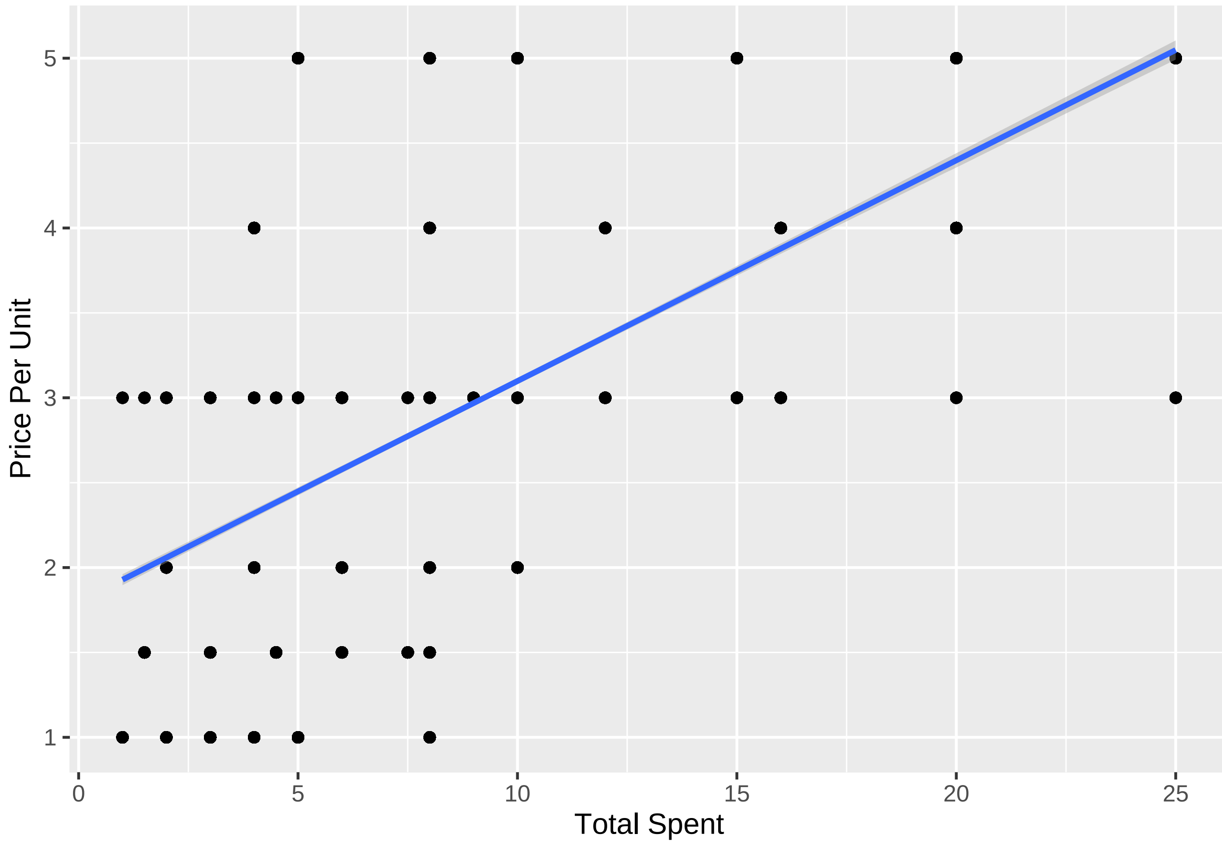

```
cafe_data_cleaned %>%
  ggplot(aes(x = `Total Spent`, y = `Price Per Unit`))+
  geom_point()
```



Modeling and Inference

```r
cafe_data_cleaned %>%
  ggplot(aes(x = `Total Spent`, y = `Price Per Unit`))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
model<- lm(Quantity~`Price Per Unit`, data = cafe_data_cleaned)

summary(model)
```

```
##
## Call:
## lm(formula = Quantity ~ `Price Per Unit`, data = cafe_data_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.04057 -1.02741 -0.02741  0.97917  1.98575
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.007668   0.035665  84.330   <2e-16 ***
## `Price Per Unit` 0.006581   0.011132   0.591    0.554
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.385 on 9998 degrees of freedom
## Multiple R-squared:  3.496e-05,  Adjusted R-squared:  -6.506e-05
## F-statistic: 0.3496 on 1 and 9998 DF,  p-value: 0.5544
```

Conclusion

This project explored the full data science workflow using the Dirty Cafe Sales dataset — a realistically messy dataset containing 10,000 café transactions. Through rigorous data cleaning, including handling missing values, correcting inconsistent formatting, and converting variables to appropriate types, we transformed a chaotic dataset into a reliable source for analysis.

We then conducted exploratory data analysis (EDA), uncovering key business insights such as the most popular items, highest revenue-generating products, and patterns in purchasing behavior. Summary statistics helped describe central trends in spending, while visualizations and group-level aggregations highlighted the café's top-performing offerings.

Overall, this project showcased the importance of data cleaning as a foundational step in the data science process, and demonstrated how even initially unreliable data can yield valuable insights when properly wrangled. These insights can be used to guide business decisions, optimize inventory, and tailor marketing strategies to customer preferences.